UNIVERSITY OF MINES AND TECHNOLOGY

TARKWA, GHANA

SCHOOL OF POSTGRADUATE STUDIES

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DEVELOPMENT OF A WEB-BASED POSTGRADUATE ADMINISTRATIVE SUPPORT SYSTEM FOR THE UNIVERSITY OF MINES AND TECHNOLOGY

A Project Report Submitted to the Department of Computer Science and Engineering,

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

The University of Mines and Technology (UMaT), located in Tarkwa, Ghana, is a premier institution renowned for its engineering and technology-oriented programmes. The School of Postgraduate Studies at UMaT oversees a growing population of graduate students across diverse disciplines including Computer Science, Mining Engineering, Environmental Science, Geological Engineering, Electrical Engineering, and Mechanical Engineering. As enrolment figures continue to rise, the administrative burden associated with managing postgraduate academic activities has intensified significantly (Addo & Mensah, 2022).

Historically, postgraduate administrative processes at UMaT have relied heavily on manual, paper-based workflows. Activities such as course registration, thesis submission tracking, grade entry, financial status verification, graduation clearance processing, and result publication have been conducted through fragmented systems involving physical forms, spreadsheets, and face-to-face interactions. This traditional approach, while functional in smaller institutional settings, has proven increasingly inadequate as the volume and complexity of postgraduate operations have grown (Asamoah & Oheneba-Sakyi, 2023).

The global trend in higher education has witnessed a paradigm shift towards digitalisation and automation of academic administrative processes. Student Information Systems (SIS), Learning Management Systems (LMS), and Enterprise Resource Planning (ERP) solutions have become integral components of modern university infrastructure (Al-Mamary et al., 2021). These systems streamline operations, reduce human error, improve data accessibility, and enhance the overall student experience. Institutions such as the University of Ghana, Kwame Nkrumah University of Science and

Technology (KNUST), and the University of Cape Coast have invested substantially in digital transformation initiatives to modernise their administrative frameworks (Owusu-Ansah et al., 2023).

Despite these advancements in the broader Ghanaian higher education landscape, many specialised postgraduate processes remain underserved by existing commercial solutions. Generic student information systems often fail to accommodate the unique workflows inherent in postgraduate administration, such as multi-stage thesis supervision cycles, role-based clearance approval chains, and Cumulative Weighted Average (CWA) computation for degree classification (Boateng & Agyemang, 2022). This gap underscores the need for a tailored, institution-specific digital solution that comprehensively addresses the administrative requirements of postgraduate studies at UMaT.

The present study therefore focuses on the development of a web-based Postgraduate Administrative Support System designed to digitise, streamline, and centralise the key administrative functions of the School of Postgraduate Studies at UMaT. The system is built upon modern web technologies, employing a React-based frontend with TypeScript for type safety, Tailwind CSS for responsive design, and a Backend-as-a-Service (BaaS) architecture leveraging cloud-hosted PostgreSQL databases and serverless edge functions for AI-powered features.

1.2 Problem Statement

The current administrative framework at the School of Postgraduate Studies, UMaT, is characterised by several critical inefficiencies that impede the smooth operation of academic and administrative processes. The reliance on manual, paper-based systems for course registration, thesis tracking, grade management, financial verification, and graduation clearance creates a fragmented operational environment where data silos, communication delays, and process bottlenecks are commonplace (Mensah & Antwi, 2023).

Specifically, the following challenges have been identified through stakeholder consultations and process analysis:

Fragmented Data Management: Student records, financial information, thesis progress data, and examination results are stored across disparate systems and physical files, making holistic data retrieval and cross-referencing extremely difficult. This fragmentation increases the likelihood of data inconsistencies, duplication, and loss (Gyamfi & Adu-Gyamfi, 2021).

Inefficient Thesis Supervision Workflow: The thesis submission and review process relies on physical document exchange between students and supervisors. There is no centralised platform for tracking submission history, managing feedback, or monitoring student progress through defined thesis milestones. This results in delays, lost feedback, and difficulty in maintaining an audit trail of supervisory interactions.

Manual Grade Processing: Grade entry, CWA calculation, and result publication are performed manually, increasing the risk of computational errors and delaying the availability of results to students. The absence of automated validation mechanisms means that data entry errors may propagate through the system undetected (Asante & Owusu, 2022).

Opaque Financial Status Tracking: Students lack real-time visibility into their fee payment status, outstanding balances, and payment history. Financial officers spend considerable time manually reconciling payment records and generating compliance reports.

Cumbersome Clearance Process: The multi-departmental graduation clearance process requires students to physically visit multiple offices to obtain sequential approvals. This process is time-consuming, prone to bottlenecks, and lacks transparency regarding the status of pending clearance requests.

These challenges collectively result in administrative delays, reduced stakeholder satisfaction, and an institutional environment that falls short of

the digital transformation standards being adopted by peer institutions. There is therefore a compelling need for an integrated, web-based system that addresses these inefficiencies comprehensively.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design and develop a web-based Postgraduate Administrative Support System that digitises and centralises the key administrative processes of the School of Postgraduate Studies at the University of Mines and Technology, Tarkwa.

1.3.2 Specific Objectives

The specific objectives of this study are to:

i. Develop a role-based access control system that supports six distinct user roles: Student, Supervisor, Administrator, Dean, Accountant, and Examinations Officer, each with tailored dashboard interfaces and functional capabilities.

ii. Implement a digital course registration module that enables postgraduate students to register for core and elective courses, with real-time credit hour tracking and academic period management.

iii. Design and implement a thesis submission and supervision management module that supports multi-stage thesis tracking (Proposal through Defence), chapter upload functionality, and an integrated feedback mechanism with AI-assisted review capabilities.

iv. Develop an automated grade entry and CWA computation system that supports both manual data entry and CSV bulk upload, with built-in validation and result publication workflows.

v. Create a multi-stage digital graduation clearance system that enables sequential departmental approvals with real-time progress tracking and status notifications.

vi. Implement comprehensive financial management features including fee status dashboards, payment processing interfaces, financial analytics, and exportable PDF reports.

vii. Integrate AI-powered supervision support tools that leverage large language models to provide automated feedback suggestions, quality checks, and a conversational assistant for thesis supervisors.

1.4 Scope of the Study

This study encompasses the design, development, and deployment of a web-based Postgraduate Administrative Support System specifically tailored for the School of Postgraduate Studies at the University of Mines and Technology. The scope of the system covers the following functional domains:

The system supports six user roles with distinct access privileges and functional interfaces: Student, Supervisor, Administrator, Dean, Accountant, and Examinations Officer. Each role has a dedicated dashboard with role-specific statistics, quick actions, and recent activity feeds.

For the Student role, the system provides course registration, thesis submission and progress tracking, academic result viewing with CWA computation and performance charts, financial status monitoring with payment capabilities, document request management, examination timetable access, and multi-stage graduation clearance tracking.

For the Supervisor role, the system offers assigned student management, thesis submission review with document preview, remark and feedback management, AI-powered feedback suggestions, and a dedicated AI conversational assistant with automated quality checking capabilities.

For administrative roles (Administrator, Dean, Accountant, and Examinations Officer), the system provides student enrollment management, fee status tracking, pass list generation with PDF export, CWA results analytics, clearance approval workflows, financial analytics with filterable charts, and grade entry with result publication.

1.4.1 Limitations

The system is designed as a prototype and does not currently integrate with existing university databases or legacy systems. Authentication is implemented as a mock system for demonstration purposes and would require integration with an institutional single sign-on (SSO) solution for production deployment. Additionally, the system does not currently support mobile native applications, although the responsive design ensures usability across mobile devices via web browsers.

1.5 Significance of the Study

This study contributes to the growing body of knowledge on the digitalisation of higher education administration in Ghana and sub-Saharan Africa. The significance of this work can be examined from multiple perspectives:

Institutional Efficiency: The system eliminates redundant manual processes, reduces paperwork, and streamlines workflows across all administrative functions. By centralising data and automating routine tasks, the system significantly reduces the time and effort required to manage postgraduate administrative activities (Asante & Owusu, 2022).

Enhanced Stakeholder Experience: Students gain real-time access to their academic records, financial status, and clearance progress, reducing the need for physical visits to administrative offices. Supervisors benefit from AI-assisted tools that enhance the quality and efficiency of thesis feedback (Hwang et al., 2023).

Data-Driven Decision Making: The system provides comprehensive analytics dashboards for administrators, deans, and accountants, enabling data-driven decision making regarding student performance, financial compliance, and resource allocation.

Technological Innovation: The integration of AI-powered features for thesis supervision represents a novel application of large language models in the context of Ghanaian postgraduate education, contributing to the discourse on AI-augmented academic support systems.

Scalability and Replicability: The modular architecture and role-based design of the system make it adaptable for deployment in other institutions with similar administrative structures, contributing to the broader digitalisation of higher education in the region.

1.6 Organisation of the Report

This report is organised into the following chapters:

Chapter 1: Introduction provides the background of the study, problem statement, objectives, scope, significance, and organisation of the report.

Chapter 2: Literature Review presents a comprehensive review of related literature, theoretical frameworks, existing systems, and identified gaps that justify the development of the proposed system.

Chapter 3: System Analysis and Design describes the analysis of the existing system, requirements specification, and the detailed design of the proposed system including use case diagrams, data flow diagrams, entity relationship diagrams, system architecture, and user interface design.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter presents a comprehensive review of the literature relevant to the development of the Postgraduate Administrative Support System. The review examines theoretical frameworks underpinning student information systems, analyses existing solutions in the domain, evaluates the technological landscape for web-based academic systems, and identifies gaps that the proposed system seeks to address. The discussion draws from academic journals, conference proceedings, textbooks, and reputable institutional reports to establish a robust theoretical and empirical foundation for the study.

2.2 Theoretical Framework

2.2.1 Student Information Systems (SIS)

Student Information Systems (SIS) are software applications designed to manage student data, including enrolment, academic records, financial information, and administrative processes. According to Aldiab et al. (2019), a well-designed SIS serves as the operational backbone of academic institutions, facilitating data-driven decision making and streamlining administrative workflows. The evolution of SIS from basic record-keeping tools to comprehensive, web-based platforms reflects the broader digital transformation of higher education (Al-Mamary et al., 2021).

In the context of postgraduate education, SIS requirements differ significantly from undergraduate systems. Postgraduate programmes involve research-intensive activities, thesis supervision cycles, and complex degree classification processes that require specialised functionality not typically found in generic SIS solutions (Aghaee & Keller, 2019). The proposed system addresses these unique requirements by incorporating thesis management,

CWA computation, and multi-stage clearance workflows that are specifically designed for postgraduate administrative contexts.

2.2.2 Role-Based Access Control (RBAC)

Role-Based Access Control (RBAC) is a widely adopted security model that restricts system access based on the roles assigned to individual users within an organisation. The National Institute of Standards and Technology (NIST) defines RBAC as a mechanism where permissions are associated with roles, and users are assigned to appropriate roles, thereby simplifying the management of access privileges (Sandhu et al., 2000). Ferraiolo et al. (2001) further elaborated on the RBAC reference model, identifying core components including users, roles, permissions, and sessions.

In the context of university administration, RBAC is particularly relevant because different stakeholders require varying levels of access to institutional data. The proposed system implements RBAC with six distinct roles: Student, Supervisor, Administrator, Dean, Accountant, and Examinations Officer. Each role is associated with a unique set of permissions and a tailored user interface, ensuring that users can only access functions and data relevant to their administrative responsibilities (Asamoah & Oheneba-Sakyi, 2023).

2.2.3 Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM), proposed by Davis (1989), provides a theoretical lens for understanding user adoption of information systems. TAM posits that two primary factors influence a user's decision to adopt a technology: perceived usefulness (the degree to which a user believes the system will enhance their performance) and perceived ease of use (the degree to which the user believes the system will be free of effort). Venkatesh and Davis (2000) extended TAM to include social influence and cognitive instrumental processes.

The design of the proposed system is informed by TAM principles. The user interface employs a clean, intuitive design with consistent navigation patterns, role-specific dashboards, and contextual quick actions to maximise perceived ease of use. Functional features such as real-time financial tracking, automated CWA computation, and AI-assisted feedback are designed to demonstrate clear utility, thereby enhancing perceived usefulness and facilitating adoption across diverse user groups (Hwang et al., 2023).

2.3 Review of Existing Systems

2.3.1 Commercial Student Information Systems

Several commercial SIS platforms dominate the higher education technology market. Oracle PeopleSoft Campus Solutions, Ellucian Banner, and Jenzabar provide comprehensive student lifecycle management capabilities (Oracle, 2023). However, these systems are primarily designed for large-scale institutions in developed countries and present significant challenges for adoption in Ghanaian universities, including high licensing costs, complex implementation requirements, and limited customisability for local administrative workflows (Owusu-Ansah et al., 2023).

SAP Student Lifecycle Management offers enterprise-grade functionality but requires substantial institutional IT infrastructure and expertise for deployment and maintenance. The cost of implementation and ongoing support often exceeds the budgetary capacity of many Ghanaian institutions, particularly for specialised postgraduate functionality (Boateng & Agyemang, 2022).

2.3.2 Open-Source Alternatives

Open-source solutions such as OpenSIS and Fedena provide more accessible alternatives for institutions with limited budgets. OpenSIS offers student demographics management, scheduling, grade book functionality, and reporting capabilities (OpenSIS, 2023). However, these platforms typically

focus on K-12 or undergraduate education and lack the specialised features required for postgraduate administration, such as thesis supervision tracking, multi-stage clearance processes, and research progress monitoring (Gyamfi & Adu-Gyamfi, 2021).

Moodle and Canvas, while excellent as Learning Management Systems, primarily address course delivery and assessment rather than the broader administrative functions that constitute the focus of the proposed system. The integration of LMS functionality with administrative process management remains a significant gap in the current educational technology landscape (Al-Mamary et al., 2021).

2.3.3 Institutional Systems in Ghanaian Universities

Within the Ghanaian context, several institutions have developed or adopted bespoke student management systems. The University of Ghana employs the Student Records and Management Information System (SRMIS) for undergraduate and postgraduate record management. KNUST utilises a combination of the Student Portal and the College Management Information System (CMIS) for student-facing services and administrative functions respectively (Owusu-Ansah et al., 2023).

However, these systems are typically designed for the specific operational contexts of their host institutions and may not be directly transferable to UMaT without substantial modification. Furthermore, many of these systems focus primarily on undergraduate operations and provide limited support for the unique requirements of postgraduate administration, particularly in areas such as thesis supervision management and research progress tracking (Mensah & Antwi, 2023).

2.4 Thesis Supervision and Management in Higher Education

The management of thesis supervision is a critical function in postgraduate education that directly impacts student completion rates and research quality.

Research by Aghaee and Keller (2019) demonstrates that digital platforms for thesis management can reduce average completion time by up to 23% through improved communication, structured milestone tracking, and timely feedback mechanisms. Wisker (2012) emphasises that effective supervision requires systematic tracking of student progress across defined stages, regular feedback provision, and clear documentation of supervisory interactions.

The proposed system addresses these requirements through a structured thesis management module that tracks student progress across seven defined stages (Proposal, Chapters 1-5, and Defence). The system enables digital submission of thesis chapters, provides document preview capabilities, maintains a chronological record of supervisor remarks and feedback, and offers AI-powered feedback suggestions to enhance the quality and consistency of supervisory input. This approach aligns with the concept of "augmented supervision" proposed by Hwang et al. (2023), where AI tools complement rather than replace human supervisory judgment.

2.5 Automated Grading and CWA Computation

The automation of grade processing and academic performance calculation has been a significant area of development in educational technology. Ihantola et al. (2010) highlight the importance of automated assessment systems in reducing administrative burden and improving the accuracy and timeliness of academic result processing. In the Ghanaian university system, the Cumulative Weighted Average (CWA) is the primary metric for degree classification, computed as the weighted mean of course marks across all semesters (Asante & Owusu, 2022).

The proposed system implements an automated grade entry module that supports both manual data input and CSV bulk upload functionality. Built-in validation ensures data integrity by verifying index number formats against established patterns (e.g., UMaT/PG/XXXX/YY) and grade values against the

institutional grading scale (A+ through F). The CWA computation algorithm calculates weighted averages based on course credit hours, providing immediate results that can be reviewed and published to student portals through a controlled workflow.

2.6 Financial Management in Higher Education

Financial transparency and efficient fee management are essential components of university administration. Johnstone and Marcucci (2010) argue that clear financial communication between institutions and students significantly reduces disputes and improves overall satisfaction. Digital financial management systems enable real-time fee tracking, automated compliance monitoring, and comprehensive reporting that supports institutional financial planning (Mensah & Antwi, 2023).

The proposed system provides students with a comprehensive financial status dashboard displaying total fees, amounts paid, outstanding balances, and payment history. The system supports multiple payment channels including Mobile Money (MTN, Vodafone Cash, AirtelTigo), bank card payments, and manual receipt upload with verification workflows. For financial administrators, the system offers analytics dashboards with filterable charts showing monthly revenue collections, compliance rates by programme, and exportable PDF reports for institutional records.

2.7 Digital Clearance Processes

Graduation clearance is a multi-departmental process that verifies a student's eligibility for degree conferment. Traditional clearance processes require students to physically visit multiple offices to obtain sequential approvals, a practice that is time-consuming and prone to bottlenecks (Asamoah & Oheneba-Sakyi, 2023). Digital clearance systems transform this process by enabling online submission of clearance requests, real-time status tracking, and electronic approval workflows.

The proposed system implements a six-stage clearance process encompassing School Fees verification, Library clearance, Departmental clearance, Thesis Submission confirmation, ICT Directorate clearance, and final Dean of Postgraduate approval. Each stage displays its current status (Cleared, Pending, or Not Started) along with relevant metadata such as the approving officer and clearance date. A visual progress bar provides students with an immediate overview of their clearance completion percentage.

2.8 AI in Academic Support Systems

The integration of Artificial Intelligence (AI) in educational contexts has gained significant momentum with the advent of large language models (LLMs). Hwang et al. (2023) propose the concept of "augmented intelligence" in education, where AI tools enhance rather than replace human capabilities. In the context of thesis supervision, AI can assist supervisors by generating preliminary feedback suggestions, performing automated quality checks on student submissions, and providing an interactive advisory interface for supervision-related queries (Zawacki-Richter et al., 2019).

The proposed system integrates AI functionality through a serverless edge function that connects to a Gemini language model. The AI module operates in three modes: (1) a conversational chat assistant that helps supervisors with evaluation criteria, feedback formulation, and progress tracking; (2) an automated feedback generator that analyses submission metadata to produce categorised suggestions across content, formatting, references, methodology, and clarity dimensions; and (3) an automated quality checker that identifies potential issues in student submissions categorised by severity level. This multi-modal AI integration represents a novel contribution to the field of technology-enhanced postgraduate supervision.

2.9 Web Technologies for Educational Systems

The selection of appropriate web technologies is critical for the development of robust, maintainable, and scalable educational applications. React, developed by Facebook (now Meta), has emerged as a leading JavaScript library for building user interfaces, offering a component-based architecture that promotes code reusability and maintainability (Gackenheimer, 2015). TypeScript extends JavaScript with static type checking, reducing runtime errors and improving code quality in large-scale applications (Bierman et al., 2014).

Tailwind CSS provides a utility-first approach to styling that enables rapid UI development with consistent design tokens and responsive design capabilities (Tailwind Labs, 2023). The Backend-as-a-Service (BaaS) paradigm, exemplified by platforms such as Supabase and Firebase, provides cloud-hosted databases, authentication services, and serverless functions that significantly reduce backend development complexity (Supabase, 2024). The proposed system leverages these technologies to create a modern, responsive, and maintainable web application that can be deployed and scaled with minimal operational overhead.

2.10 Identified Gaps and Justification

The review of existing literature and systems reveals several significant gaps that the proposed system addresses:

First, there is a notable absence of integrated, postgraduate-specific administrative systems designed for the Ghanaian university context. Existing commercial solutions are prohibitively expensive and lack customisation for local workflows, while open-source alternatives focus primarily on undergraduate or K-12 education (Owusu-Ansah et al., 2023; Gyamfi & Adu-Gyamfi, 2021).

Second, the integration of AI-powered supervision support tools within a postgraduate administrative system is a novel approach that addresses the growing need for technology-enhanced research supervision. While individual AI tools for academic writing exist, their integration within a holistic administrative platform that supports the full spectrum of postgraduate operations is unprecedented in the Ghanaian educational technology landscape (Hwang et al., 2023).

Third, existing systems rarely provide comprehensive, multi-role support within a single platform. The proposed system consolidates the needs of six distinct user roles into a unified application with tailored interfaces, eliminating the need for multiple disconnected tools and improving operational cohesion (Boateng & Agyemang, 2022).

Fourth, the combination of real-time financial tracking, automated CWA computation, digital clearance workflows, and AI-assisted thesis supervision within a single platform represents a comprehensive solution that addresses the full lifecycle of postgraduate administration. This holistic approach distinguishes the proposed system from existing solutions that typically address only a subset of these functional requirements.

2.11 Summary

This chapter has reviewed the theoretical foundations, existing systems, and technological landscape relevant to the development of a Postgraduate Administrative Support System. The review has established that while significant advancements have been made in educational technology, there remains a substantial gap in the provision of integrated, postgraduate-specific administrative systems for Ghanaian universities. The proposed system addresses this gap by combining role-based access control, digital thesis management, automated academic processing, financial management, digital clearance workflows, and AI-powered supervision support within a unified, modern web application. The following chapter presents the detailed system

analysis and design that translates these requirements into a comprehensive technical solution.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter presents the comprehensive analysis and design of the Postgraduate Administrative Support System. It begins with an analysis of the existing manual processes, identifies their limitations, and proceeds to describe the proposed system in detail. The chapter covers functional and non-functional requirements, and presents the system design through use case diagrams, data flow diagrams, entity relationship diagrams, system architecture, database design, and user interface specifications.

3.2 Analysis of the Existing System

The current administrative processes at the School of Postgraduate Studies, UMaT, are predominantly manual and paper-based. Through stakeholder interviews and process observation, the following workflow characteristics were documented:

Course Registration: Students complete paper-based registration forms that are manually processed by departmental administrators. Credit hour calculations are performed manually, and registration records are maintained in spreadsheet files. This process typically takes 3-5 working days per student.

Thesis Supervision: Students submit printed copies of thesis chapters to their supervisors. Feedback is provided through handwritten annotations or verbal discussions. There is no centralised system for tracking submission history, managing milestones, or maintaining a record of supervisory interactions.

Grade Processing: Examinations officers manually enter grades into spreadsheets, compute CWA values using manual calculations or basic spreadsheet formulas, and publish results through departmental notice boards or email distributions.

Financial Management: Fee records are maintained by the Finance Office in separate ledgers and spreadsheets. Students must physically visit the Finance Office to verify their payment status, and financial reports are generated manually at the end of each academic period.

Graduation Clearance: Students physically visit six departments in sequence to obtain clearance stamps and signatures. Each department maintains its own clearance records, and there is no centralised tracking mechanism for monitoring overall clearance progress.

3.2.1 Limitations of the Existing System

The existing system suffers from several critical limitations:

i. Data fragmentation across multiple physical locations and digital files, leading to inconsistencies and difficulties in cross-referencing.

ii. High susceptibility to human error in manual data entry, grade computation, and financial record-keeping.

iii. Significant time delays in administrative processing due to sequential, paper-based workflows.

iv. Lack of real-time visibility into academic, financial, and clearance statuses for students and administrators.

v. Absence of audit trails for supervisory interactions, administrative decisions, and process changes.

vi. Limited analytical capability for institutional planning and decision-making.

vii. Physical presence requirements that create bottlenecks and inconvenience for all stakeholders.

3.3 Description of the Proposed System

The proposed Postgraduate Administrative Support System is a web-based application that digitises and centralises all key administrative functions of the School of Postgraduate Studies. The system employs a modern client-side architecture with a React-based frontend communicating with cloud-hosted backend services. It supports six user roles, each with a dedicated dashboard and tailored functional modules.

The system is designed around a modular architecture where each functional domain (course registration, thesis management, grade processing, financial management, clearance processing, and AI-assisted supervision) is implemented as a discrete module with well-defined interfaces. This modular approach facilitates maintainability, scalability, and the potential for incremental feature additions.

3.4 Requirements Specification

3.4.1 Functional Requirements

ID	Module	Requirement	User Role
FR01	Authentication	The system shall authenticate users using email-based role identification	All
FR02	Course Registration	Students shall be able to register for core and elective courses with real-time credit tracking	Student
FR03	Thesis Management	Students shall upload thesis chapters with progress tracking across 7 stages	Student
FR04	Thesis Review	Supervisors shall review submissions, preview documents, and add remarks	Supervisor
FR05	AI Feedback	The system shall generate AI-powered feedback suggestions for thesis submissions	Supervisor
FR06	Grade Entry	Exams officers shall enter grades manually or via CSV upload with validation	Exams Officer
FR0	CWA Computation	The system shall automatically	Exams Officer

7		compute CWA from validated grade entries	
FR08	Result Publication	The system shall publish approved results to student portals in batches	Exams Officer
FR09	Financial Dashboard	Students shall view total fees, payments, balances, and payment history	Student
FR10	Payment Processing	The system shall support online payment and receipt upload with verification	Student
FR11	Clearance Tracking	Students shall track multi-stage graduation clearance progress	Student
FR12	Clearance Approvals	Dean shall approve or reject clearance requests with department filtering	Dean
FR13	Student Management	Administrators shall enroll, search, and manage student records	Admin
FR14	Analytics Dashboard	The system shall provide role-specific analytics with filterable charts	Admin/Dean/ Accountant
FR15	PDF Export	The system shall generate branded PDF reports for pass lists and financial data	Admin/ Accountant/ Exams
FR16	AI Chat Assistant	Supervisors shall interact with an AI chatbot for supervision queries	Supervisor

3.4.2 Non-Functional Requirements

ID	Category	Requirement
NFR01	Performance	The system shall load dashboard pages within 2 seconds under normal network conditions
NFR02	Responsiveness	The UI shall adapt seamlessly to desktop, tablet, and mobile viewports
NFR03	Usability	The system shall provide intuitive navigation with no more than 3 clicks to reach any function
NFR04	Security	Role-based access control shall prevent unauthorised access to restricted functions and data
NFR05	Reliability	The system shall maintain 99.5% uptime with graceful error handling and fallback mechanisms
NFR06	Scalability	The architecture shall support concurrent access by at least 500 users without performance degradation
NF	Maintainability	The codebase shall follow component-based

R07		architecture with TypeScript for type safety
NF R08	Data Integrity	Grade entry shall include validation rules to prevent invalid data from being processed

3.5 System Design

3.5.1 Use Case Diagram

The use case diagram in Figure 3.1 illustrates the interactions between the six system actors (Student, Supervisor, Administrator, Dean, Accountant, and Examinations Officer) and the primary system functions. Each actor is associated with specific use cases that reflect their role-based permissions and functional requirements.

[USE CASE DIAGRAM]

The Student actor interacts with seven primary use cases: Register Courses, Upload Thesis Chapter, View Results, View Financial Status, Make Payment, Request Documents, and Track Clearance. The Supervisor actor interacts with four use cases: View Assigned Students, Review Submissions, Add Remarks, and Use AI Assistant. The Administrator actor manages student records, fee statuses, pass lists, and analytics. The Dean approves clearances and reviews CWA results. The Accountant accesses fee analytics and exports financial reports. The Examinations Officer enters grades, generates pass lists, and publishes results.

Figure 3.1: Use Case Diagram for the Postgraduate Administrative Support System

3.5.2 Data Flow Diagram

The Data Flow Diagram (DFD) in Figure 3.2 presents a Level 0 (Context Diagram) showing the system boundary and data flows between external entities and the system.

[CONTEXT DIAGRAM / DFD LEVEL 0]

At Level 0, six external entities (Student, Supervisor, Admin, Dean, Accountant, and Exams Officer) interact with the central system. Key data flows include: course registration data and thesis submissions from students; feedback, remarks, and AI queries from supervisors; student records and enrollment data from administrators; clearance decisions from the dean; financial queries from the accountant; and grade data from the examinations officer. The system processes these inputs and returns results, notifications, reports, and status updates to the respective entities.

Figure 3.2: Level 0 Data Flow Diagram (Context Diagram)

Figure 3.3 presents the Level 1 DFD, decomposing the system into its major processing subsystems.

[DFD LEVEL 1]

The Level 1 DFD identifies six primary processes: (1) Course Registration Management, which receives registration requests and updates the course database; (2) Thesis Supervision Management, which processes thesis submissions, routes them to supervisors, and manages feedback cycles; (3) Grade and Results Processing, which handles grade entry, CWA computation, and result publication; (4) Financial Management, which processes fee payments, verifies receipts, and generates financial reports; (5) Clearance Processing, which manages sequential departmental approvals and tracks clearance status; and (6) AI Processing, which interfaces with the external AI gateway to generate feedback suggestions and process chat queries.

Figure 3.3: Level 1 Data Flow Diagram

3.5.3 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram in Figure 3.4 illustrates the data entities, their attributes, and the relationships between them in the system database.

[ENTITY RELATIONSHIP DIAGRAM]

The ERD comprises the following primary entities and their relationships:

User: Central entity with attributes including user_id (PK), email, name, role, department, program, and index_number. A user can have one role (Student, Supervisor, Admin, Dean, Accountant, or Examinations Officer).

Course: Represents academic courses with attributes code (PK), name, credits, type (core/elective), and semester. A student registers for multiple courses (many-to-many relationship through a Registration entity).

Thesis Submission: Records thesis chapter uploads with submission_id (PK), student_id (FK), chapter_name, submission_date, status, and file_url. A student has multiple submissions (one-to-many).

Remark: Captures supervisor feedback with remark_id (PK), submission_id (FK), author, text, and date. A submission can have multiple remarks (one-to-many).

Grade: Stores academic grades with grade_id (PK), student_id (FK), course_id (FK), grade_value, marks, and semester. Used for CWA computation.

Fee Record: Tracks financial transactions with fee_id (PK), student_id (FK), semester, amount, paid, balance, and status.

Clearance: Manages graduation clearance with clearance_id (PK), student_id (FK), department, status, cleared_by, and cleared_date.

Figure 3.4: Entity Relationship Diagram

3.5.4 System Architecture

The system architecture, illustrated in Figure 3.5, follows a modern three-tier architecture pattern adapted for cloud-based deployment.

[SYSTEM ARCHITECTURE DIAGRAM]

The architecture comprises three primary tiers:

Presentation Tier (Frontend): Built with React 18, TypeScript, and Tailwind CSS, the frontend application runs entirely in the user's web browser. It employs a single-page application (SPA) architecture with React Router for

client-side navigation and TanStack React Query for server state management. The UI layer uses the shadcn/ui component library built on Radix UI primitives, ensuring accessibility compliance and consistent design language.

Application Tier (Backend Services): The backend leverages a Backend-as-a-Service (BaaS) architecture with cloud-hosted PostgreSQL for data persistence and Deno-based edge functions for serverless compute. The supervisor-ai edge function interfaces with the Lovable AI Gateway to access the Google Gemini language model for AI-powered features. Authentication is managed through JWT-based token verification.

Data Tier (Database): A PostgreSQL database hosted on cloud infrastructure stores all persistent data. The database schema supports the entities defined in the ERD with appropriate foreign key relationships, indexes, and constraints. Row-Level Security (RLS) policies can be configured to enforce data access controls at the database level.

Figure 3.5: System Architecture Diagram

3.6 Database Design

The database schema is designed to support the full range of administrative functions implemented by the system. Table 3.1 summarises the primary database entities and their key attributes.

Table 3.1: Database Schema Summary

Entity	Key Attributes	Relationships
Users	id, email, name, role, department, program, index_number, avatar_url	Has many: Submissions, Grades, Fees, Clearances
Courses	code, name, credits, type, semester	Many-to-many with Users via Registrations
Registrations	id, student_id, course_code, semester, status	FK to Users, FK to Courses

Submissions	id, student_id, chapter, date, status, file_url	FK to Users; Has many Remarks
Remarks	id, submission_id, text, date, author	FK to Submissions
Grades	id, student_id, course_code, grade, marks, semester	FK to Users, FK to Courses
Fees	id, student_id, semester, amount, paid, balance, status	FK to Users
Clearances	id, student_id, department, status, cleared_by, date	FK to Users
Result Batches	id, semester, year, department, program, student_count, status	References Grades

3.7 User Interface Design

The user interface design follows a consistent layout pattern across all user roles, comprising three primary regions: a fixed sidebar navigation panel, a top header bar with user information and actions, and a main content area. The design system employs semantic colour tokens defined through CSS custom properties, ensuring visual consistency and supporting both light and dark display modes.

3.7.1 Login Interface

The login interface features a split-screen layout with the institutional branding panel on the left (displaying the UMaT logo, system name, and university motto) and the authentication form on the right. The form captures email and password credentials and routes users to their role-specific dashboards upon successful authentication. On mobile devices, the layout adapts to a single-column format with the branding information displayed above the login form.

3.7.2 Dashboard Interface

Each user role has a dedicated dashboard displaying four role-specific statistic cards, a recent activity feed, and a quick actions panel. For example, the

Student dashboard displays registered courses count, thesis progress, CWA, and pending reviews. The Supervisor dashboard shows assigned student count, pending reviews, approvals this month, and average review time. This role-specific customisation ensures that each user sees the most relevant information immediately upon login.

3.7.3 Student Module Interfaces

The Course Registration interface presents courses in two sections (Core and Elective) with tabular layouts on desktop and card-based layouts on mobile. Real-time credit hour and course count tracking is displayed in summary cards above the course listings.

The Thesis Submission interface features a visual progress tracker showing the current stage across seven milestones, a drag-and-drop file upload area, and a submission history panel displaying previous uploads with their approval statuses.

The Results interface provides a comprehensive academic performance view with an overall CWA display, semester-by-semester CWA cards with trend indicators, a bar chart for CWA performance analysis, and detailed grade tables for each semester.

The Clearance interface displays a visual progress bar indicating overall clearance completion percentage, followed by individual clearance step cards showing department name, description, status badge, and relevant metadata.

3.7.4 Supervisor Module Interfaces

The Assigned Students interface presents a summary statistics panel and a detailed table listing all supervised students with their index numbers, programmes, current thesis stages, and progress statuses.

The Review Submissions interface provides a list view of all submissions (pending and reviewed) and a detailed review view featuring a document preview panel, a remark input area, previous remarks display, and an optional

AI feedback sidebar that can be toggled on or off. The AI panel displays categorised feedback suggestions with options to use, copy, or rate each suggestion.

3.7.5 Administrative Module Interfaces

The Manage Students interface includes a search bar, an enrollment form modal, and a data table with columns for name, index number, programme, department, status, and actions. The Grade Entry interface supports both manual row-by-row data entry and CSV file upload, with real-time validation indicators and CWA computation capabilities.

The Fee Analytics interface provides filterable bar charts for monthly revenue collections and compliance rates by programme, with summary statistics cards for total collected, outstanding amounts, compliance rates, and defaulter counts. All financial interfaces support multi-dimensional filtering by academic year and department.

3.8 Summary

This chapter has presented a comprehensive analysis and design of the Postgraduate Administrative Support System. The analysis of the existing manual system revealed significant inefficiencies in data management, thesis supervision, grade processing, financial tracking, and clearance administration. The proposed system addresses these limitations through a modular, web-based architecture that supports six distinct user roles with tailored interfaces and functionalities. The system design has been documented through use case diagrams, data flow diagrams, entity relationship diagrams, and a detailed system architecture specification. The database schema and user interface designs provide the foundation for the implementation phase of the project.

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